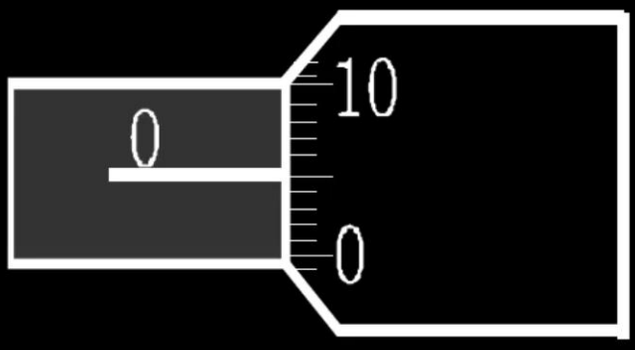
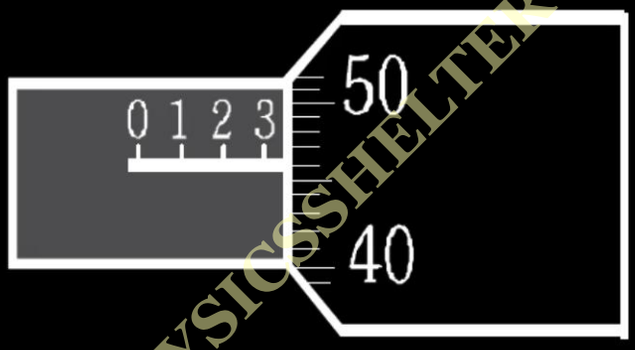
EXAMPLE:

The circular scale of a micrometer has 200 divisions and pitch of 2 mm. Find the measured value of thickness of a thin sheet (In mm).





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Solution: Here since pitch is 2mm, there is no need to make 1mm etching marks on the main scale (∵ 1mm would correspond to half rotation of the circular scale). ∴ main scale etchings must be 2 mm apart.

Now: l.c. = 2mm/200 = 0.01 mm.

Zero error must be +ve ∵ reference line is above zero of circular scale. (Tightening would have made the zero coincide with reference line). ∴ ZERO ERROR = 5 × 0.01 = +0.05 mm (∵ fifth division was coinciding).

Now in second figure, third etch is visible ⇒ main scale reading = 3 × 2 = 6mm, Circular scale reading = 46 × 0.01 = 0.46 mm.

⇒ Reading = 6.46mm ⇒ length = 6.46 − 0.05 = 6.41mm Ans.

EXAMPLE :

In a screw gauge, there are 100 division in circular scale and each main scale division is of 1 mm. When there is no gap between the jaws, 97th division coincides with the main scale zero and zero of main scale is not visible. While measuring the diameter of a ball, the circular scale is between 3 mm mark and 4 mm mark such that the 76th division of circular scale coincides with the reference line. Select the correct alternative :

(A) The least count of the micrometer is 0.01 cm

(B) The zero error is –0.04 mm

(C) The diameter of the ball is 3.79 mm

(D) There is positive zero error in the screw guage

Solution: Pitch = 1mm, L.C. = 1mm/100 = 0.01 mm.

∵ Zero of main scale is invisible, zero error must be negative.

Now zero error = −1mm + 97×0.01 mm = −0.03 mm.

Reading = 3mm + 76×0.01 = 3.76 mm.

⇒ Correct diameter = 3.76 − (−0.03) = 3.79 mm Ans. (C)

EXAMPLE :

A screw gauge having 100 equal divisions and a pitch of length 1 mm is used to measure the diameter of a wire of length 5.6 cm. The main scale reading is 1 mm and 47th circular division coincides with the main scale. Find the curved surface area of the wire in cm^2 to appropriate significant figures.(Use $\pi = 22/7$)

Solution: least count = 1mm/100 = 0.01 mm.

⇒ Diameter of wire = 1mm + 47×0.01 mm = 1.47 mm.

Now Curved surface area = $2\pi r l = \pi d l$

⇒ $CSA = \frac{22}{7} \times 0.147 \times 5.6 = 2.5872$, Now round to 2 Sig figs

(∵ 5.6 cm has 2 significant figures) ⇒ $CSA = 2.6\ cm^2$

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EXAMPLE:

In the determination of Young's modulus $\left(Y = \frac{4MLg}{\pi l d^2}\right)$ by using Searle's method, a wire of length L = 2 m and diameter d = 0.5 mm is used. For a load M = 2.5 kg, an extension l = 0.25 mm in the length of the wire is observed. Quantities d and l are measured using a screw gauge and a micrometer, respectively. They have the same pitch of 0.5 mm. The number of divisions on their circular scale is 100. The contributions to the maximum probable error of the Y measurement is (2011)

(a) due to the errors in the measurements of d and *l* are the same

(b) due to the error in the measurement of d is twice that due to the error in the measurement of *l*

(c) due to the error in the measurement of *l* is twice that due to the error in the measurement of *d*

(d) due to the error in the measurement of d is four times that due to the error in the measurement of *l*

Solution: By taking log and differentiating. $\frac{\Delta Y}{Y} = \frac{\Delta M}{M} + \frac{\Delta L}{L} + \frac{\Delta g}{g} + \frac{\Delta l}{l} + 2\frac{\Delta d}{d}$

Least count of screw gauge and micrometer = 0.5/100 = 0.005 mm.

Now error due to l measurement: $\frac{\Delta l}{l} = \frac{0.005}{0.25} = 0.02$

Error due to d measurement: $2\frac{\Delta d}{d} = \frac{2 \times 0.005}{0.5} = 0.02$

Thus both have equal error contributions (option A).

EXAMPLE:

The density of a solid ball is to be determined in an experiment. The diameter of the ball is measured with a screw gauge, whose pitch is 0.5 mm and there are 50 divisions on the circular scale. The reading on the main scale is 2.5 mm and that on the circular scale is 20 divisions. If the measured mass of the ball has a relative error of 2% the relative percentage error in the density is (2011)

(a) 0.9%

(b) 2.4%

(c) 3.1%

(d) 4.2%

Solution: $\rho = \frac{m}{4/3\pi(d/2)^3} \Rightarrow \frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + \frac{3\Delta d}{d}$,

l.c. = 0.5/50 = 0.01 mm, $d = 2.5 + 20 \times 0.01 = 2.70$ mm

$\Rightarrow \frac{\Delta\rho}{\rho} = 2\% + 3\times\left(\frac{0.01}{2.70}\right)\times 100\% = 3.11\% \approx 3.1\%$ (C) Ans.

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EXAMPLE:

Q 5. In a screw gauge, the zero of main scale coincides with fifth division of circular scale as shown in the figure (i). The circular scale of a screw gauge has 50 divisions and its pitch is 0.5 mm. The diameter of the ball being measured in the figure (ii) is _____ (2006)

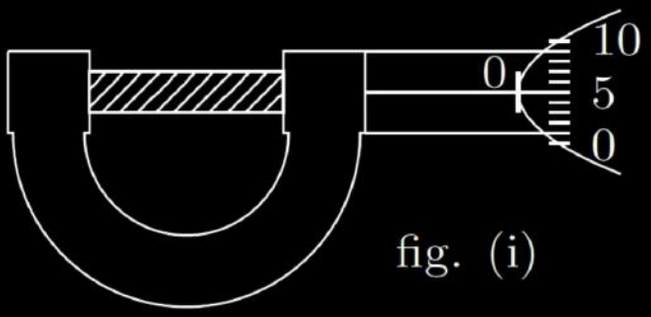


fig. (i)

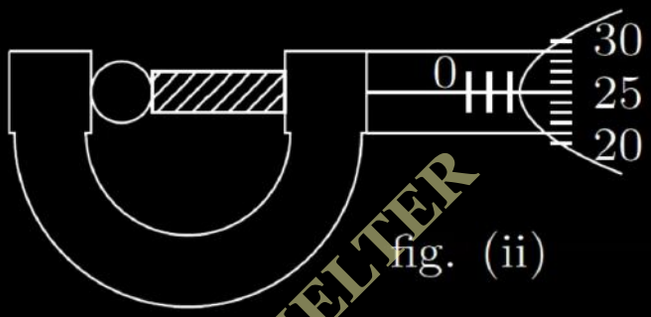


fig. (ii)

Solution: ∵ Reference line is above zero of circular scal, zero error is positive. Now l.c. = 0.5/50 = 0.01 mm. From second figure, main scale reading = 0.5 × 3 = 1.5 mm. (Unless stated otherwise, etchings on main scale are taken as equal to pitch).

Reading : 1.0 + 25 × 0.01 mm = 1.25 mm. (∵ Two divisions on main scale = 1mm)

Zero Error : 5 × 0.01 = + 0.05 mm

⇒ Correct length = 1.25 − 0.05 = 1.20 mm Ans.

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EXAMPLE:

Q 10. Consider a Vernier callipers in which each 1 cm on the main scale is divided into 8 equal divisions and a screw gauge with 100 divisions on its circular scale. In the Vernier callipers, 5 divisions of the Vernier scale coincide with 4 divisions on the main scale and in the screw gauge, one complete rotation of the circular scale moves it by two divisions on the linear scale. Then,

(2015)

(A) If the pitch of the screw gauge is twice the least count of the Vernier callipers, the least count of the screw gauge is 0.01 mm.

(B) If the pitch of the screw gauge is twice the least count of the Vernier callipers, the least count of the screw gauge is 0.005 mm.

(C) If the least count of the linear scale of the screw gauge is twice the least count of the Vernier callipers, the least count of the screw gauge is 0.01 mm.

(D) If the least count of the linear scale of the screw gauge is twice the least count of the Vernier callipers, the least count of the screw gauge is 0.005 mm.

Solution: $MSD = \frac{1}{8}\,cm = 1.25\,mm$, $5\,VSD = 4\,MSD \Rightarrow VSD = 1\,mm$

$\Rightarrow$ l.c. of Vernier $= 0.25\,mm. = lcv$

Let each linear scale division on screw gauge be $m \Rightarrow p = 2m$

$\Rightarrow$ l.c. of screw gauge $= p/100 = 2m/100. = lcs$

(A) If $p = 2\,lcv = 2 \times 0.25\,mm = 0.5\,mm \Rightarrow lcs = \frac{0.5}{100} = 0.005\,mm$

A is incorrect

(B) is correct (from analysis for part A).

(C) If $m = 2\,lcv = 0.5\,mm \Rightarrow lcs = 2m/100 = 0.01\,mm$

$\Rightarrow$ C is correct.

(D) is wrong (from analysis for part C)

Correct options B & C.

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